The empirical recurrence for OEIS A317227, recovered from its tail

Adrian Perez Fontelles
Independent researcher

8 September 2026

Abstract

OEIS A317227 records "Empirical recurrence of order 84" and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 139$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 5$, so the recurrence is proved for every $n \geq 89$. Its last coefficient is $c_{84} = 13312 \neq 0$, so no further tail satisfies anything shorter; any order-84 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry's recurrence is the one displayed here, whatever its linked file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture that is not written down

OEIS A317227 is "Number of n X 5 0..1 arrays with every element unequal to 0, 1, 2, 3, 6, 7 or 8 king-move adjacent elements, with upper left element zero.". It has offset 1 and begins

$$16, 120, 488, 2306, 17039, 101712, 575421, 3632635, \ \ldots$$

The entry states:

> Empirical recurrence of order 84 (see link above).

As of the "Last modified" line on the live entry (Oct 19 2021 21:52:08, revision 6) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry's condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u, and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) \ = \ \iota^{\top} M^{\,j(n)} \tau$$

for a linear reindexing j fixed by the entry's shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 139$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 139$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S'+1$ consecutive values of a. ☐

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 84: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 5$ is the first whose minimal order is exactly the stated 84.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 278$ exact terms from $n = 5$ on, it returns order 84 — the order the entry states — with integer coefficients:

$$a(n) \;=\; \sum_{i=1}^{84} c_i\, a(n-i),$$

c_1	11	c_{22}	−182818821	c_{43}	−3029548948	c_{64}	−1334551778
c_2	−37	c_{23}	1656711106	c_{44}	2135141134	c_{65}	306375851
c_3	138	c_{24}	−907351105	c_{45}	−6677778072	c_{66}	265075419
c_4	−995	c_{25}	−167045705	c_{46}	−3701670006	c_{67}	616894691
c_5	3046	c_{26}	−3948366721	c_{47}	−4276269851	c_{68}	9653830
c_6	−6024	c_{27}	2111869835	c_{48}	7563000225	c_{69}	−154688059
c_7	31845	c_{28}	1520829804	c_{49}	6378590108	c_{70}	−210270435
c_8	−96645	c_{29}	6847840954	c_{50}	4068763225	c_{71}	−61626444
c_9	137114	c_{30}	−3427171066	c_{51}	−6360074372	c_{72}	70017220
c_{10}	−517765	c_{31}	−3998189410	c_{52}	−5122818438	c_{73}	49227998
c_{11}	1596157	c_{32}	−8838419275	c_{53}	−2568659650	c_{74}	27455235
c_{12}	−1869797	c_{33}	3053707292	c_{54}	3673604494	c_{75}	−21052900
c_{13}	5132587	c_{34}	7190627614	c_{55}	3352922298	c_{76}	−7649246
c_{14}	−16136285	c_{35}	8870466380	c_{56}	810583203	c_{77}	−5774140
c_{15}	15622374	c_{36}	−389827899	c_{57}	−1480376846	c_{78}	3880912
c_{16}	−31628895	c_{37}	−9804299988	c_{58}	−2673002045	c_{79}	707760
c_{17}	107907511	c_{38}	−6469544749	c_{59}	418046084	c_{80}	572544
c_{18}	−83124162	c_{39}	−2717129833	c_{60}	645904249	c_{81}	−394688
c_{19}	113661319	c_{40}	8849208509	c_{61}	2156760163	c_{82}	−15616
c_{20}	−501110660	c_{41}	2121084355	c_{62}	−685557152	c_{83}	−19968
c_{21}	309219062	c_{42}	4943902781	c_{63}	−416295257	c_{84}	13312

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 89$, and it is the recurrence OEIS A317227 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 5$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{84} - c_1 x^{84-1} - \cdots - c_{84}$. Its constant term is $-c_{84} = -13312$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 84 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 84, so $\deg q = 84$, and it is monic. It holds from some index t on. From $\max(t, 5)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 84, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 19 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry's English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 84, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry's.

Third, the recovered recurrence was evaluated directly on the entry's own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 13312.

Remark 4. The conjecture's own statement was never read. The entry pins it down to the family of order-84 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A317227.

[2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.

[3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)

[4] M. Kauers and P. Paule, *The Concrete Tetrahedron*, Springer, 2011. (C-finite sequences and their closure properties.)

[5] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.